
A Comparative Review of LLM Quantization Methods (2020-2025)

Assignment 1 — Comparative Literature Review

INF503 / AI503 Machine Learning

Mohamed Elrashid · 22002576

Supervisor: Dr. Manar Alkhatib

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Introduction

BACKGROUND

- By 2025, LLMs moved from research curiosities to **industrial integration**.
- Llama 3 scaled to **405B parameters** trained on **15.6T tokens** ([RP20]).
- A 70B model at 16-bit needs over **140 GB of VRAM** to load its weights ([RP12]).
- That requirement excludes almost all consumer-grade hardware and inflates cloud TCO.

WHY THIS MATTERS

- Compression is no longer a peripheral optimization. It is a prerequisite for commercial viability.
- The **cloud-scale training vs. edge deployment gap** is the central deployment problem of 2025.

BRIEF OVERVIEW OF THE AREA

- Anchor methods cut memory at 4-bit: GPTQ ([RP09]), SmoothQuant ([RP10]), AWQ ([RP15]).
- The frontier has moved to **sub-4-bit**: QuIP# ([RP21]) and AQLM ([RP30]) push 2-bit.
- Yet findings sit scattered across NeurIPS, ICML, ICLR, MLSys, and arXiv.
- This work brings that literature into one structured view via a **50-paper SLR**.

Problem Statement

THE "SCATTERED KNOWLEDGE" ISSUE

- AI research has outpaced the industry's ability to standardize deployment.
- Sub-fields differ in **target** (weight-only, W+A, KV-cache), **bit width**, and **calibration recipe**.
- Comparing GPTQ ([RP09]) to QuaRot ([RP29]) to KIVI ([RP38]) is not obvious from paper titles.

WHY IT MATTERS

- Practitioners face a hard search task when picking a method for given hardware and accuracy.
- Researchers struggle to spot the open problems.
- Enterprise AI ROI suffers from this fragmentation.

CURRENT CHALLENGES — THE EVALUATION MONOCULTURE

- Benchmarking is mostly done on **data-centre A100s / H100s**, far from real edge constraints.
- Convergence on **WikiText-2 perplexity** and **Llama** as the de facto test bed.
- Many "SOTA" claims do not translate to consumer hardware.

Field rhetoric says edge. Field practice says A100.

Research Objectives

Aim. Turn the fragmented quantization literature into a navigable taxonomy that supports both researchers and practitioners.

SPECIFIC OBJECTIVES

O1 · Reproducible Collection. Gather at least 50 papers (2016–2025) using ResearchRabbit citation tracing and PRISMA-style screening.

O2 · Comparative Analysis. Map every paper across **8 standardized dimensions**: Year, Paper, Contribution, Dataset, Data Size, Models, Evaluation Metrics, Key Results.

O3 · Contribution Taxonomy. Organize the field into three buckets: **Compression Methods, Behavioural Effects, Foundational Context.**

O4 · Trend and Gap Synthesis. Surface systemic patterns and the evaluation blind spots that block research-to-production transitions.

BEYOND AN ACADEMIC SUMMARY

This research delivers a direct critique of the field's trajectory. The output highlights exactly where industrial readiness is being compromised.

Research Gaps and Deployment Risks

The field is **methods-mature** but **evaluation-narrow**. Gaps are not just missing data points. They are unmeasured business risks for any edge deployment.

Gap dimension	Missing	Strategic deployment risk
Energy / power reporting	94%	Unpredictable OPEX and thermal failure on mobile
Edge / mobile testing	94%	Lab-to-field gap; A100 benchmarks mislead ROI
Safety / trustworthiness	80%	Alignment drift; quantized models may bypass filters
Long-context evaluation	76%	RAG and long-document tasks fail in production

RARE EXCEPTIONS WORTH NAMING

- **Safety after compression:** [RP26], [RP41], [RP50]
- **Edge hardware:** AWQ ([RP15]), SpinQuant ([RP36]), LLM in a flash ([RP37])

WHY THIS MATTERS

Without Joules-per-token or safety-drift numbers, enterprise deployment of quantized LLMs is a **gamble, not a calculation**.

Future Work — Toward Honest Evaluation

The next generation of LLM efficiency must move past simple bit reduction. Three priorities for the research agenda:

1. EDGE-QUANTIZATION BENCHMARKS

- Mandate testing on real target hardware: **Jetson modules, M-series Macs, mobile NPUs.**
- Shift the primary efficiency metric to **Joules per token.**
- Enables accurate power-budgeting for mobile and IoT deployment.

2. SAFETY-SCREEN MINI-SUITES

- Integrate **HarmBench** ([RP44]) or **OpenMiniSafety** ([RP50]) subsets as a default pipeline step.
- Catches post-compression alignment drift in under thirty minutes.

3. ADVANCED LONG-CONTEXT PROBES

- Adopt **LongBench**, **RULER**, and **Needle-in-Haystack** as default tests.
- Verifies that KV-cache compression ([RP22], [RP38]) does not break model memory.

WHAT THIS CHANGES

These are **evaluation defaults**, not research breakthroughs. Adopting them moves the field from academic curiosity to deployment-grade infrastructure.

Proposed Next Step — From Gap to Experiment

A1 surfaced the **safety-after-quantization** gap. The proposed next step is to take the smallest testable slice of that gap and run it as an empirical study.

PRE-REGISTERED HYPOTHESIS (H1)

*At NF4 (4-bit), the relative drop in **adversarial-refusal rate** is expected to exceed the relative drop in **MMLU accuracy**, both measured against the FP16 baseline of the same model.*

PROPOSED EXPERIMENTAL SETUP

- **Model:** Qwen2.5-1.5B-Instruct (small enough for edge deployment).
- **Precisions to compare:** FP16 (baseline), INT8, NF4.
- **Safety probes:** ~550 prompts from **AdvBench**, **HarmBench**, and **XSTest** (false-refusal).
- **Utility probes:** 1,000-question MMLU subset and WikiText-2 perplexity.
- **Hardware:** free-tier Colab GPU, full pipeline targeted to run end-to-end in under 90 minutes.

PRE-DECLARED DECISION RULE

Outcome	Decision
$\Delta\text{Refusal}$ exceeds ΔMMLU by ≥ 5 pp on any safety benchmark	H1 supported
Both deltas within 5 pp of each other	Inconclusive
$\Delta\text{MMLU} \geq \Delta\text{Refusal}$ on every safety benchmark	H1 rejected

The decision rule is fixed **before** running the experiment, to remove the temptation of post-hoc reasoning.

Conclusion

THE TECHNICAL TRAJECTORY

- The 4-bit floor of 2020 dropped to credible **2-bit** by 2024.
- Made possible by **outlier-flattening rotations** ([RP29], [RP36]) and **lattice codebooks** ([RP21], [RP30]).
- **W4A4KV4** became a near-standard with **~99% zero-shot retention** at 70B ([RP29]).

THE BOTTLENECK OF HONESTY

- We can compress a 70B model to fit on consumer hardware.
- Yet we know little about its **power, safety, and long-range reasoning** in the field.
- The **cost side** of compression has matured. The **benefit side** has not.

Final remarks

The future of LLMs depends on evaluation infrastructure that reflects deployment reality.

The next research step belongs not to new compression tricks. It belongs to methods that take **honest evaluation** seriously.

Only by shifting focus from *how to compress* to *how to evaluate honestly* can the next generation of LLMs be as reliable as it is efficient.

Thank you

Questions are welcome.